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ECE 2202 – HW 2

1. A device can be modeled using a voltage source in series with a resistance. That device is shown in Figure P3.2.

a. Draw the circuit elements inside the device

A set of resistors was connected to the terminals of the device, one at a time, and the voltage at the device terminals, v_t , was measured in each case. The results are shown in Table P3.1, where R is the resistor value attached to the device. Then, a $10[\Omega]$ resistor was connected to the terminals of the device.

- b. Using KVL and Ohm's law write an equation that includes v_t and i_T
- c. Find the power absorbed by this $10[\Omega]$ resistor, when it is connected to the device terminals.

(PWA Problem 4 in Module 1)

Table P3.1

R in $[\Omega]$	v_t in $[V]$
5	-3.32
15	-24.2
20	-113

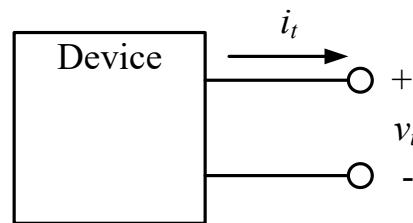
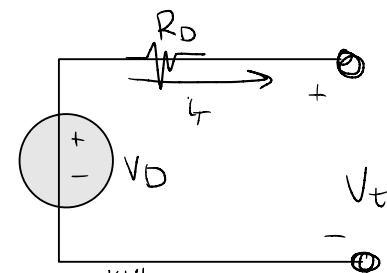


Figure P3.2



$$\textcircled{1} \quad -V_D + i_T R_D + v_t = 0$$

$$v_t = V_P = -i_T R_D$$

$$\textcircled{2} \quad \text{KVL} \quad -V_D + V_R + v_t = 0$$

$$\uparrow$$

$$V_R = i_T R_D \quad \leftarrow V_D = i_T R_D + v_t$$

$$i_T = \frac{v_t}{R}$$

$$V_D = \frac{v_t}{R} R_D + v_t$$

$$V_{D1}: \frac{-3.32[V]}{5[\Omega]} R_D - 3.32[V] = V_D \quad R_D = -21.99[\Omega]$$

$$V_D = 11.2843[V]$$

$$V_{D2}: \frac{-24.2[V]}{15[\Omega]} R_D - 24.2[V] = V_D$$

$$V_D = i_T R_D$$

$$11.2843[V] = i_T (-21.99[\Omega]) \Rightarrow i_T = \frac{11.2843[V]}{-21.99[\Omega]} = -0.513[A]$$

$$\textcircled{3} \quad P_{abs} = i_T^2 (10[\Omega])$$

$$= (-0.513[A])^2 \cdot 10[\Omega]$$

$$= 2.633[W]$$

2. A device can be modeled using a current source in parallel with a resistor. This device was connected to a 5[V] voltage source, as shown in Figure P3.3 a), and a current i_5 of 0.93[A] resulted. The same device was then connected to a 4[A] current source as shown in Figure P3.3 b), and a voltage v_4 of -34[V] resulted.

Find the voltage v_3 if a 3[Ω] resistor, as shown in Figure P3.3 c), is connected to the same device.

(PWA Problem 5 in Module 1)

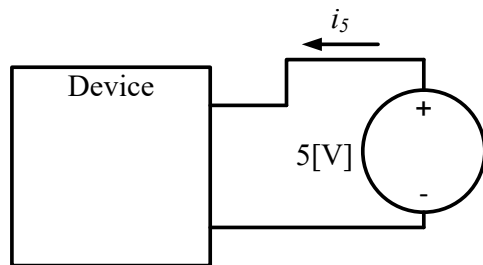


Figure P3.3 a)

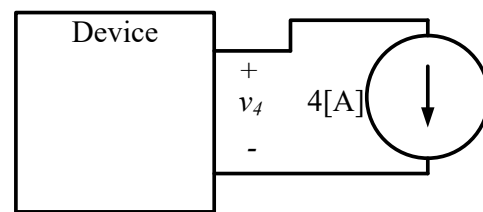


Figure P3.3 b)

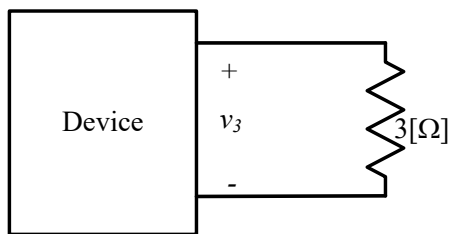
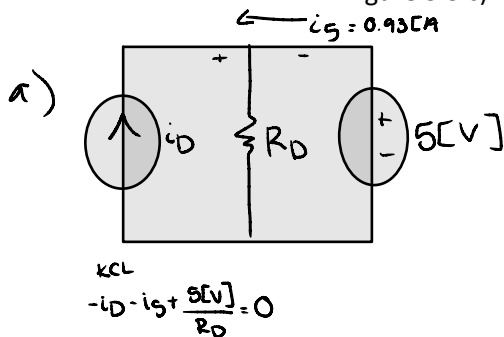


Figure 3.3 c)



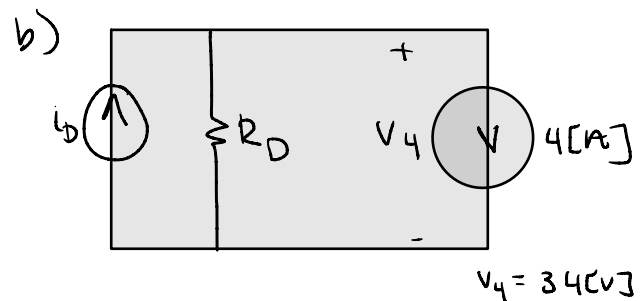
eqn a + b will solve for R_D + i_D

$$4[A] - i_D - \frac{34[V]}{R_D} = 0$$

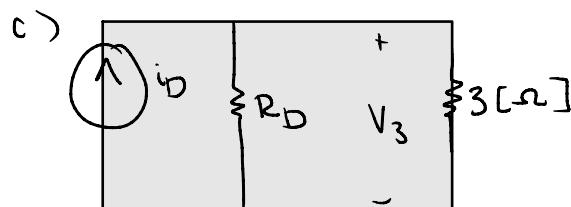
$$-0.93[A] + \frac{5[V]}{R_D} - i_D = 0$$

$$i_D = -0.3[A]$$

$$R_D = 7.9[\Omega]$$



$$4[A] - i_D + \frac{34[V]}{R_D} = 0$$



$$-i_D + \frac{v_3}{R_D} + \frac{v_3}{3[\Omega]} = 0$$

$$-(-0.3[A]) + \frac{v_3}{7.91[\Omega]} + \frac{v_3}{3[\Omega]} = 0$$

$$v_3 = -0.67[V]$$

3. A device can be modeled by a voltage source in series with a resistance. This device has the relationship between voltage v_D and current i_D shown in Figure P3.4a, using the reference polarities shown in Figure P3.4b. This relationship holds for all values of v_D and i_D . When this device is connected to a 12[V] voltage source as shown in Figure P3.4c, find the power absorbed by the device.

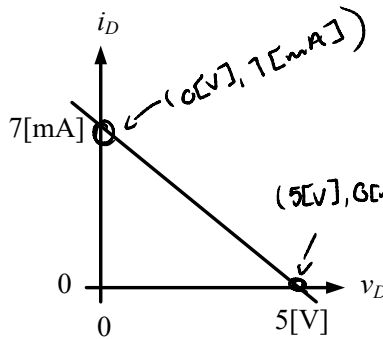


Figure P3.4a

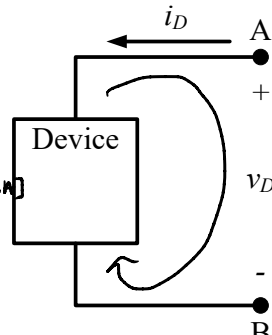


Figure P3.4b KVL

$$-V_{Th} - i_D R_{eq} + V_D = 0$$

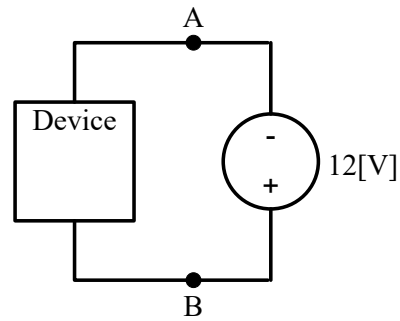
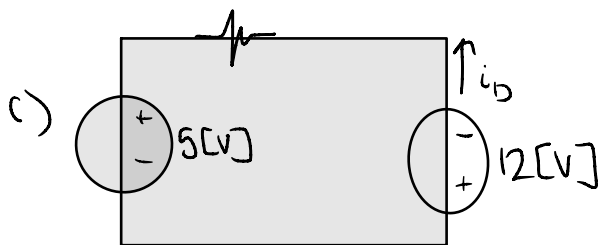


Figure P3.4c

V_{Th} & R_{eq}

$$\begin{aligned} \text{KCL form} \quad 1) \quad V_{Th} + 7[mA] + \frac{0V}{R_{eq}} &= 0 \rightarrow -V_{Th} + 7[mA]R_{eq} = 0 & R_{eq} &= \frac{5[V]}{7[mA]} = -714.286[\Omega] \\ 2) \quad V_{Th} + 0[mA] + \frac{5[V]}{R_{eq}} &= 0 & -V_{Th} + 5[V] &= 0 & V_{Th} &= 5[V] \end{aligned}$$



Find i_D

KVL

$$-5[V] - i_D(-714.28[\Omega]) - 12[V] = 0$$

$$i_D(714.28[\Omega]) = 17[V]$$

$$i_D = 23.8[mA]$$

$$\begin{aligned} P_{Abs, 12[V]} &= -(23.8[mA]) \cdot 12[V] \\ &= -0.285[W] \end{aligned}$$

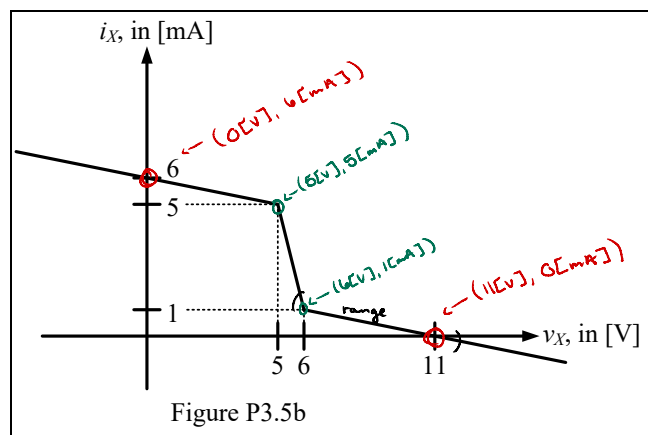
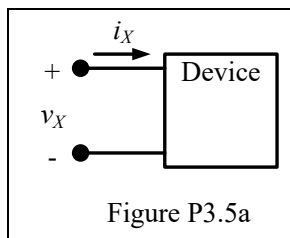
we need to figure out the
Power_{Abs, 12[V]} = $-i_D \cdot 12[V]$
↑
active

4. A device, shown in Figure P3.5a, can be modeled by a current source in parallel with a resistance. The relationship between the current through the device, i_X , and the voltage across the device, v_X , is given in the plot in Figure P3.5b.

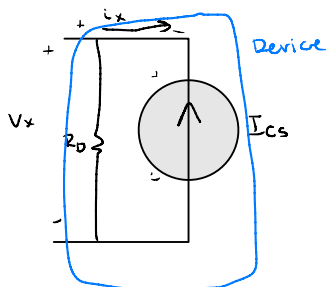
a) Find a model for the device that would be valid when current is in the range

$1[\text{mA}] < i_X < 5[\text{mA}]$. This model must have numerical values for the current and resistance, and the polarities with respect to v_X and i_X should be shown in a diagram.

b) A voltage source is applied across the device so that $v_X = 10[\text{V}]$. Find the power delivered by the device in this situation.



a)



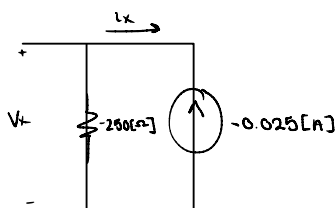
KCL:

$$-i_X - I_{CS} + \frac{v_X}{R_D} = 0$$

$v_X + i_X$ use (5, 5)
(6, 1)

① $-5[\text{mA}] - I_{CS} + \frac{5[\text{V}]}{R_D} = 0$ $I_{CS} = -0.025[\text{A}]$
 $R_D = -250[\Omega]$

② $-1[\text{mA}] - I_{CS} + \frac{6[\text{V}]}{R_D} = 0$

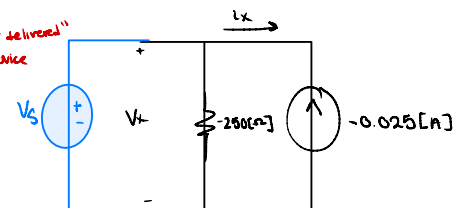


b)

P_{abs}	1V	-1V
P_{del}	-1V	1V

"Power absorbed = Power delivered"
by voltage source by device

$v_X = 10[\text{V}] = V_S$



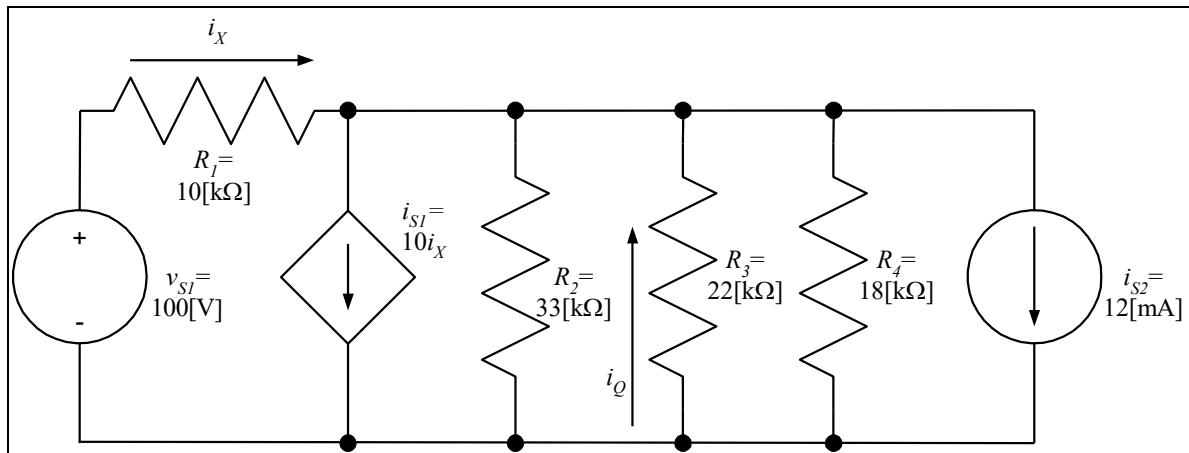
$P_{abs, 10[\text{V}]} = V \cdot I = [10[\text{V}] \cdot i_X]$

KCL: $i_X + (-0.025[\text{A}]) = \frac{10[\text{V}]}{-250[\Omega]}$
 $i_X = 0.065[\text{A}]$

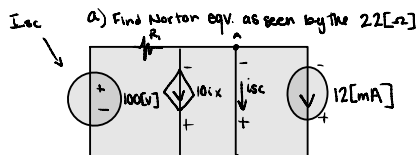
$P_{abs, 10[\text{V}]} = 10[\text{V}] \cdot 0.065[\text{A}] = 0.65[\text{W}]$

$P_{del, 10[\text{V}]} = -10[\text{V}] \cdot 0.065[\text{A}]$
 $-0.65[\text{W}]$

5. a) Find the Norton equivalent as seen by the $22[\text{k}\Omega]$ resistor.
 b) Attach the Norton equivalent that you found, to the $22[\text{k}\Omega]$ resistor. Use this circuit to solve for i_Q .



PEQWS Module 4 Problem 3



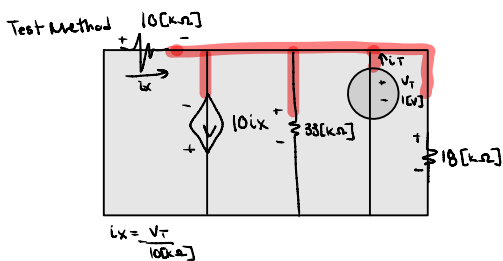
Note: when you short the terminals A+B, R_1 and R_2 will get shorted out.

$$i_X = \frac{10[\text{V}]}{10[\text{k}\Omega]} = 10[\text{mA}]$$

KCL:

$$i_{SC} + 10i_X - 10[\text{mA}] + 12[\text{mA}] = 0$$

$$i_{SC} = -0.102[\text{A}]$$



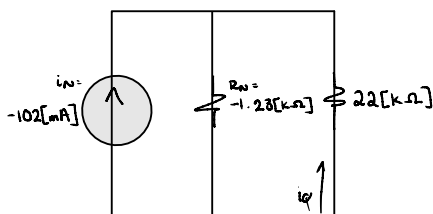
KCL

$$-i_T + \frac{V_T}{33[\text{k}\Omega]} + 10i_X - i_X + \frac{V_T}{18[\text{k}\Omega]} = 0$$

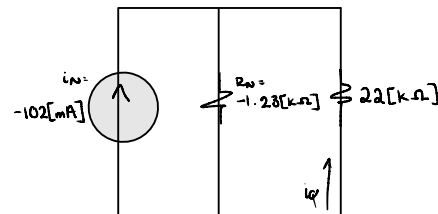
$$i_T = -0.03[\text{mA}] + 10(-0.10[\text{mA}]) - (-0.10[\text{mA}]) + 0.036[\text{mA}]$$

$$i_T = -814[\mu\text{A}]$$

$$R_N = R_{eq} = \frac{V_T}{i_T} = \frac{1[\text{V}]}{-814[\mu\text{A}]} \rightarrow -1.23[\text{k}\Omega]$$



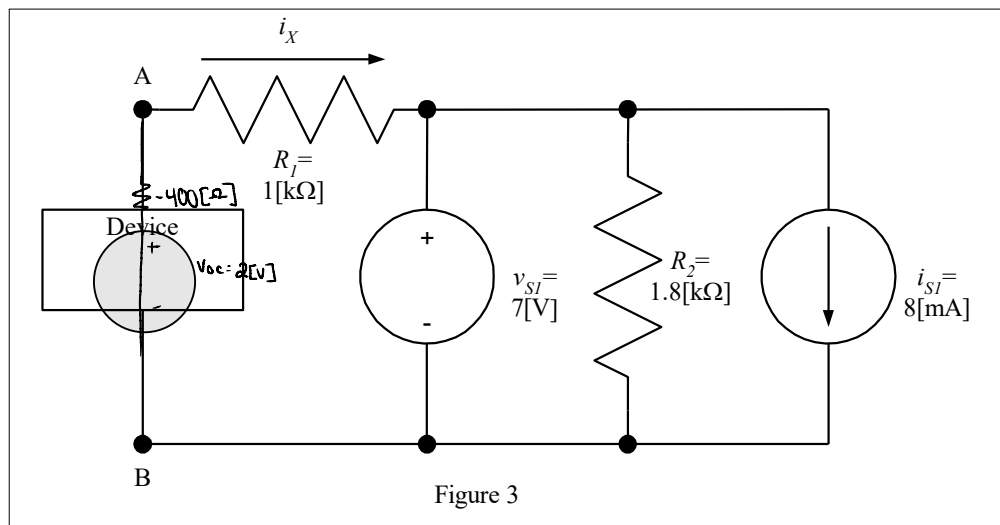
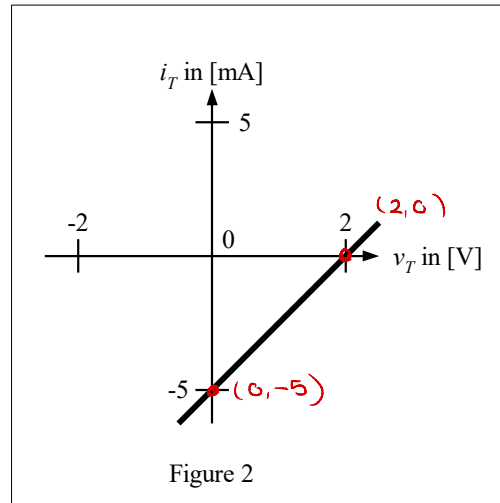
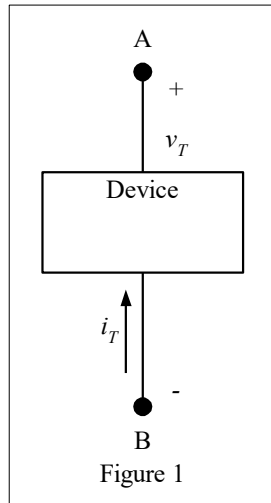
b)



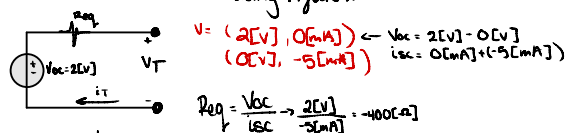
$$i_Q = -i_N \left(\frac{R_N}{R_N + 22[\text{k}\Omega]} \right)$$

$$-102[\text{mA}] \left(\frac{-1.23[\text{k}\Omega]}{22[\text{k}\Omega] - 1.23[\text{k}\Omega]} \right) = -6.03[\text{mA}]$$

6. The device in Figure 1 can be modeled with a voltage source in series with a resistance. The current and voltage for the device are related as shown in the plot in Figure 2. The device has been connected in a circuit shown in Figure 3. Find i_X . [PEQWS Module 4 Problem 4](#)



Solve for the Device using Figure 2.



this double checks our work.

$$V_T - V_{oc} + i_T R_{eq} \rightarrow V_T = 0 \text{ or } i_T = -5[mA]$$

$$i_T = \frac{-V_T + V_{oc}}{R_{eq}} \quad V_{oc} = i_T R_{eq} = 2[V]$$

$$\text{slope} = m = \frac{-1}{R_{eq}}$$

$$m = \frac{-1}{-400} = 2.5[\Omega]$$

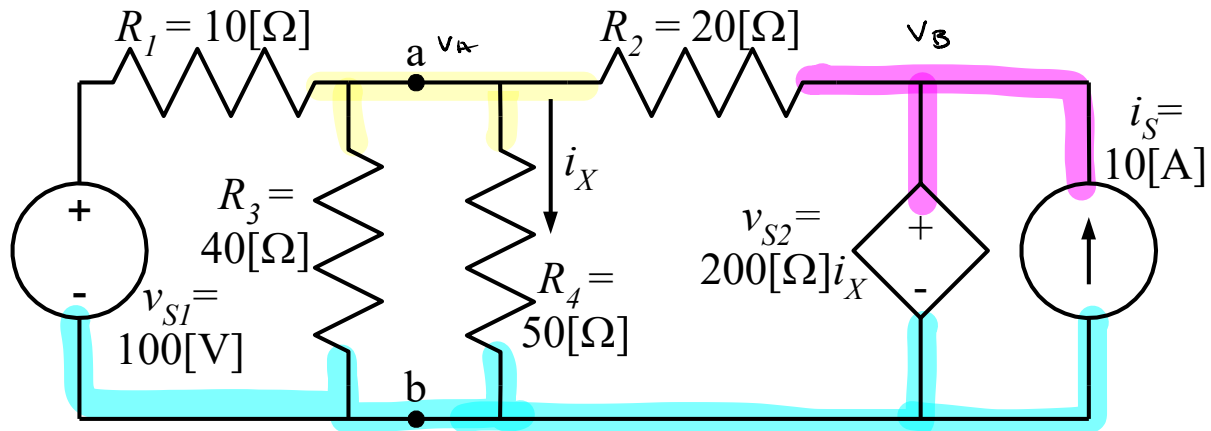
Final i_X

$$i_X = \frac{V_{oc} - v_{s1}}{R_{eq} + 1[k\Omega]} = \frac{2[V] - 7[V]}{-400 + 1[k\Omega]} = -8.33[mA]$$

7. For the circuit shown, find the Thévenin equivalent circuit as seen with respect to terminals a and b.

Draw the equivalent circuit that you found.

On this equivalent circuit you have drawn, show the values for the circuit components, and label terminals a and b.



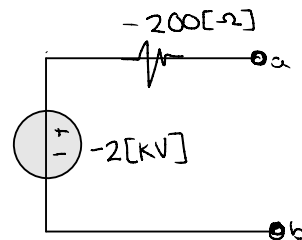
$$V_a \rightarrow \frac{V_a - 100[V]}{10[\Omega]} + \frac{V_a}{40[\Omega]} + \frac{V_a}{50[\Omega]} + \frac{V_a - V_b}{20[\Omega]} = 0$$

$$i_X = \frac{V_a}{50[\Omega]}$$

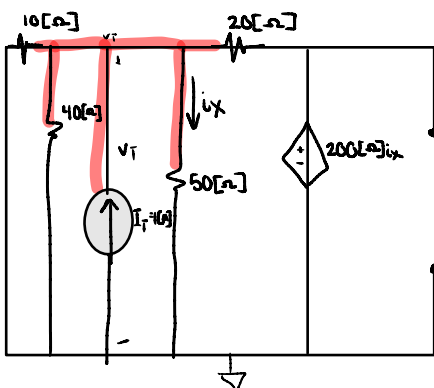
$$V_B = 200 i_X \rightarrow 200 \left(\frac{V_a}{50[\Omega]} \right) \rightarrow 4V_a$$

$$\left(\frac{V_a - 100[V]}{10[\Omega]} + \frac{V_a}{40[\Omega]} + \frac{V_a}{50[\Omega]} + \frac{V_a - 4V_a}{20[\Omega]} \right)$$

$$V_a = -2000[V]$$



To get R_{eq} , you will have to use test method cause there is a dependent source



$$\frac{V_T}{10[\Omega]} + \frac{V_T}{40[\Omega]} - 1[A] + \frac{V_T}{50[\Omega]} + \frac{V_T - 200[\Omega] i_X}{20[\Omega]} = 0$$

$$i_X = \frac{V_T}{50[\Omega]}$$

$$V_T = -200[V]$$

$$i_X = -4[A]$$

$$R_{eq} = \frac{V_T}{i_T} = \frac{-200[V]}{1[A]} = -200[\Omega]$$

Solutions

1. $p_{ABS,10/\Omega} = 8.851 \text{ [W]}$
2. $v_3 = -0.6481 \text{ [V]}$
3. Solution omitted
4. Solution omitted
5. Solution omitted
6. $i_X = -0.3787 \text{ [A]}$
7. $v_{TH} = -2000 \text{ [V]}$ (sign depends on polarity of source with respect to terminals), $R_{TH} = -200 \text{ [\Omega]}$